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Strict reciprocity for Schubert varieties in the Lagrangian Grassmannian

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Abstract. Let X_κ be the Schubert variety in the Lagrangian Grassmannian $\mathrm{LG}(m,2m)$ indexed by a strict partition κ, and $H_\kappa(j) = \chi(X_\kappa, \mathcal{O}(j))$ its Hilbert polynomial. We show that $(-1)^{|\kappa|}H_\kappa(-j)$ counts strictly order-preserving maps from the shifted diagram of κ to $\{1,\dots,2j-1\}$ that are even on diagonal boxes. Equivalently, H_κ is the Ehrhart polynomial of the order polytope of the shifted diagram in the lattice that is $\mathbb{Z}$ on diagonal and $\frac{1}{2}\mathbb{Z}$ on off-diagonal coordinates. The factor $\frac{1}{2}$ is the bond 2 of short-root steps in Lakshmibai–Seshadri paths, and the proof combines the path model with Ehrhart–Macdonald reciprocity. Peeling the top two values off the strict counts gives a recursion whose coefficients agree, up to sign, with Buch and Ravikumar's K-theoretic Pieri coefficients of the Schubert divisor; odd values play the role of their primed letters. Consequences: the negative integer zeros of H_κ are exactly $-1,\dots,-\lceil r/2\rceil$, with r the length of a longest chain, and the zeros and signs of H_κ at negative half-integers are determined completely.

Keywords: Lagrangian Grassmannian, Schubert varieties, shifted diagrams, Ehrhart reciprocity, order polytopes, Lakshmibai–Seshadri paths, K-theoretic Pieri rule

1 Introduction

The negative values of a Hilbert polynomial $H(j) = \chi(X, L^j)$ are where duality lives; for Cohen–Macaulay X one has $(-1)^{\dim X}H(-j) = \chi(X, \omega_X \otimes L^j)$. For Schubert varieties in a minuscule G/P this is combinatorics. The coordinate ring degenerates to the Hibi ring of a distributive lattice $J(P)$ [10, 9, 18], the Hilbert function is Stanley's order polynomial $j \mapsto \Omega(P, j+1)$, and Stanley's reciprocity $\Omega(P,-n) = (-1)^{|P|}\overline{\Omega}(P,n)$ [19, 21] turns negative values into counts of strictly order-preserving maps.

The Lagrangian Grassmannian $\mathrm{LG}(m,2m) = \mathrm{Sp}_{2m}/P_m$ is cominuscule but not minuscule, and this picture fails at once. Its Schubert varieties X_κ are indexed by strict partitions with parts at most m, and the relevant poset is the shifted diagram, exactly as for the orthogonal Grassmannian $\mathrm{OG}(m,2m+1)$; but the Hilbert functions differ. For $\kappa = (2,1)$, $X_\kappa = \mathrm{LG}(2,4)$ is a 3-dimensional quadric with $H(j) = (j+1)(j+2)(2j+3)/6$, while the

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order polynomial of the 3-chain $P_{(2,1)}$ is $\binom{j+3}{3}$, the Hilbert function of $\mathrm{OG}(2,5) \cong \mathbb{P}^3$. We determine what replaces Stanley reciprocity on LG.

For a strict partition $\kappa = (\kappa_1 > \cdots > \kappa_\ell > 0)$, the shifted diagram $D(\kappa)$ is the set of boxes (a,b) with $0 \le a < \ell$ and $a \le b \le a + \kappa_{a+1} - 1$. We write P_κ for $D(\kappa)$ with the product order. The boxes (a,a) are the *diagonal boxes*. Let $r(\kappa)$ be the number of elements of a longest chain in P_κ; $r(\kappa) = \max_i(\kappa_i + 2i - 2)$. Let $X_\kappa \subseteq \mathrm{LG}(m,2m)$ be the Schubert variety of dimension $|\kappa|$ and $H_\kappa(j) = \chi(X_\kappa, \mathcal{O}(j))$, with $\mathcal{O}(1)$ the ample generator of the Picard group (§2).

Theorem 1.1 (Strict reciprocity on LG)**.** *Let κ be a strict partition with parts at most m.*

(a) For every integer $j \ge 0$, the value $H_\kappa(j)$ is the number of order-preserving maps $g\colon P_\kappa \to \{0,1,\dots,2j\}$ with $g(x)$ even for all diagonal boxes x.

(b) For every integer $j \ge 1$, $(-1)^{|\kappa|}H_\kappa(-j)$ is the number of strictly order-preserving maps $f\colon P_\kappa \to \{1,\dots,2j-1\}$ with $f(x)$ even for every diagonal box x.

(c) Equivalently, $H_\kappa(j) = \#\big(j\,\mathcal{O}(P_\kappa) \cap \Lambda\big)$ for all j, where $\mathcal{O}(P_\kappa) \subseteq \mathbb{R}^{P_\kappa}$ is the order polytope and Λ is the lattice of vectors whose diagonal coordinates lie in $\mathbb{Z}$ and whose other coordinates lie in $\frac{1}{2}\mathbb{Z}$.

Example 1.2. Take $\kappa = (2,1)$, $m = 2$. The poset P_κ is the chain $(0,0) < (0,1) < (1,1)$, with both ends diagonal. At $j = 3$ the only strictly increasing triple $x < y < z$ in $\{1,\dots,5\}$ with x,z even is $(2,3,4)$; at $j = 4$ there are 5; at $j = 1,2$ there are none. This matches $H(-1) = H(-2) = 0$, $H(-3) = -1$, $H(-4) = -5$.

For $\mathrm{OG}(m,2m+1)$, with Schubert varieties X_κ^{OG} of dimension $|\kappa|$ and the spinor line bundle as $\mathcal{O}(1)$, the analogue is classical.

Theorem 1.3 (Strict reciprocity on OG)**.** *For every strict κ with parts at most m, $H_\kappa^{\mathrm{OG}}(j) = \Omega(P_\kappa, j+1)$ for all j. For $j \ge 1$, $(-1)^{|\kappa|}H_\kappa^{\mathrm{OG}}(-j) = \overline{\Omega}(P_\kappa, j-1)$, the number of strictly order-preserving maps $P_\kappa \to \{1,\dots,j-1\}$.*

This is a composite of minuscule standard monomial theory [12, 18, 9, 10] and Stanley reciprocity; we claim no novelty for it. Remark 4.5 sketches a K-theoretic proof, using [3, Lemma 4.5], that runs parallel to the LG case.

Theorem 1.4 (The integer window)**.** *Put $W(\kappa) = \lceil r(\kappa)/2 \rceil$ on* LG *and $W(\kappa) = r(\kappa)$ on* OG*, and $s_j(\kappa) = (-1)^{|\kappa|}H_\kappa(-j)$ (resp. with H_κ^{OG}). Then*

$$0 = s_1(\kappa) = \cdots = s_W(\kappa) < s_{W+1}(\kappa) \le s_{W+2}(\kappa) \le \cdots .$$

So the zero set of H_κ on the negative integers is exactly $\{-1,\dots,-W(\kappa)\}$, and $\operatorname{sgn} H_\kappa(-j) = (-1)^{|\kappa|}$ for $j > W(\kappa)$.

Since H_κ is a polynomial, it can be evaluated at half-integers, and these values are Euler characteristics of vector bundles: $2^m H_\kappa(-a-\frac{1}{2}) = \chi(X_\kappa, \bigwedge^\bullet U \otimes \mathcal{O}(-a))$, with U the tautological subbundle (§6). Write $\kappa' = (\kappa_2, \kappa_3, \dots)$ and $N_\kappa(j)$ for the count in Theorem 1.1(b), with $N_\varnothing \equiv 1$.

Theorem 1.5 (Half-integer values)**.** *Let $\kappa \neq \varnothing$. Then $H_\kappa(-\frac{1}{2}) = 2^{-\kappa_1}$, and for every integer $a \geq 1$:*

(i) $H_\kappa(-a-\frac{1}{2}) = 0$ *if and only if* $\kappa' \neq \varnothing$ *and* $a \leq \lceil r(\kappa')/2 \rceil$;

(ii) otherwise, $\operatorname{sgn} H_\kappa(-a-\frac{1}{2}) = (-1)^{|\kappa|-\kappa_1}$ *if* $2a \leq \kappa_1$ *and* $(-1)^{|\kappa|}$ *if* $2a \geq \kappa_1$;

(iii) $|H_\kappa(-a-\frac{1}{2})| \geq 2^{-\kappa_1} N_{\kappa'}(a)$, *with equality when* $2a \leq \kappa_1$, *and* $2^{\kappa_1} H_\kappa(-a-\frac{1}{2}) \in \mathbb{Z}$.

Corollary 1.6. *For every odd $M \geq 2m+1$, the number $(-1)^{|\kappa|} 2^m H_\kappa(-M/2)$ is a positive integer. In particular H_κ has no zero at half-integers $\leq -m-\frac{1}{2}$.*

Why the factor two. In an LS path of shape $j\omega_m$, a direction may change at time a only if a times the *bond* of each Bruhat cover crossed is an integer. On LG, adding an off-diagonal box is a reflection in a short root, with bond 2, and adding a diagonal box is a reflection in the long root $2\varepsilon_m$, with bond 1 (Lemma 3.1). So off-diagonal coordinates live in $\frac{1}{2}\mathbb{Z}$ and diagonal ones in $\mathbb{Z}$: this is Λ. Ehrhart–Macdonald reciprocity exchanges the polytope (a) with its interior (b), and integrality on the diagonal becomes the parity condition. On OG every bond is 1. On the K-theoretic side, the odd values play the primed letters of Buch and Ravikumar's Pieri rule [3], which may not sit on the diagonal (Remark 4.4).

Relation to the literature. De Concini's symplectic tableaux [6] and Littelmann's LS paths [14] give bases of $H^0(X_\kappa, \mathcal{O}(j))$, and part (a) follows quickly from the latter; it may be known to experts.[1] Chirivì's semi-toric degenerations [4] and Chirivì, Costa Cesari, Fang and Littelmann [5] relate LS paths on chains with prescribed bonds to lecture hall polytopes; we did not find there a statement about $\chi(X_\kappa, \mathcal{O}(-j))$ on LG. Stembridge's enriched P-partitions [22] satisfy a self-reciprocity, a different statement. To the best of our knowledge, Theorem 1.1(b),(c) and Theorem 1.5 are new. Full proofs will appear in a longer version.

[1] R. Dehy [7], as reported by Mathieu [17, §4], uses LS paths to interpret Demazure module dimensions as Ehrhart polynomials of a simplicial structure on $\{u \leq w\}$. We have not read [7] first-hand; as far as Mathieu's summary shows, it contains neither Λ nor part (b).

2 Preliminaries

Shifted diagrams. Let $\Delta_m = \{(a,b) \in \mathbb{Z}^2 : 0 \le a \le b \le m-1\}$ with the product order; $x \le y$ means x is weakly north-west of y. The map $\kappa \mapsto D(\kappa)$ is a bijection from strict partitions with parts at most m to order ideals of Δ_m, and $\rho \subseteq \kappa$ iff $D(\rho) \subseteq D(\kappa)$. So the up-sets of P_κ are exactly the sets $\theta = D(\kappa) \setminus D(\rho)$, $\rho \subseteq \kappa$. The coordinate sum increases strictly along chains, and the staircase $(0,0) < (0,1) < (1,1) < \cdots < (a,a) < \cdots < (a,b)$ is a chain of $a+b+1$ boxes below (a,b); hence $r(\kappa) = 1 + \max\{a+b : (a,b) \in D(\kappa)\}$. The rows ≥ 1 of P_κ form a copy of $P_{\kappa'}$ via $(a,b) \mapsto (a-1, b-1)$, preserving diagonal boxes.

Counting functions. For a finite poset P and $n \ge 0$, let $\Omega(P,n)$ (resp. $\overline{\Omega}(P,n)$) be the number of order-preserving (resp. strictly order-preserving) maps $P \to [n] = \{1,\dots,n\}$. For strict $\nu \subseteq \kappa$ and $j \ge 1$ let $N_{\kappa/\nu}(j)$ be the number of strictly order-preserving $f\colon D(\kappa) \setminus D(\nu) \to [2j-1]$ that are even on diagonal boxes, and $N_\kappa = N_{\kappa/\varnothing}$. For $N \ge 0$ let $F_\kappa(N)$ be the number of order-preserving $g\colon P_\kappa \to \{0,\dots,N\}$ that are even on diagonal boxes.

Lemma 2.1. *Let $\kappa \ne \varnothing$ and $r = r(\kappa)$.*

(a) $N_\kappa(j) > 0$ iff $j \ge \lceil r/2 \rceil + 1$, and $N_\kappa(j) \le N_\kappa(j+1)$ for all $j \ge 1$.

(b) $\overline{\Omega}(P_\kappa, n) > 0$ iff $n \ge r$, and $\overline{\Omega}(P_\kappa, n) \le \overline{\Omega}(P_\kappa, n+1)$.

Proof. (a) $f(a,b) = a+b+2$ is strictly order-preserving with values in $[2, r+1]$ and even on the diagonal. Conversely, along the staircase chain $f(0,0) \ge 2$ (it is even) and f increases strictly over r boxes, so $\max f \ge r+1$. Hence $N_\kappa(j) > 0$ iff $2j-1 \ge r+1$; as $2j-1$ is odd, this is $j \ge \lceil r/2 \rceil + 1$. Monotonicity: $[2j-1] \subseteq [2j+1]$ preserves parity. (b) is the same with $f(a,b) = a+b+1$ and no parity. □

Reciprocity. If $Q \subseteq \mathbb{R}^d$ is a d-polytope with vertices in a lattice Γ of rank d, then $L_Q(t) = \#(tQ \cap \Gamma)$ is a polynomial of degree d and $L_Q(-t) = (-1)^d \#(\mathrm{int}(tQ) \cap \Gamma)$ for $t \ge 1$ (Ehrhart–Macdonald reciprocity [16, 1]). The order polytope $\mathcal{O}(P) = \{y \in [0,1]^P : x \le x' \Rightarrow y_x \le y_{x'}\}$ is full-dimensional with vertices the indicator vectors of up-sets [20]; its interior is cut out by $0 < y_x < 1$ and $y_x < y_{x'}$ for $x < x'$.

Schubert varieties and cited inputs. Let $G = \mathrm{Sp}_{2m}$ with simple roots $\alpha_i = \varepsilon_i - \varepsilon_{i+1}$ ($i < m$), $\alpha_m = 2\varepsilon_m$, and $\omega = \omega_m = \varepsilon_1 + \cdots + \varepsilon_m$. Then $G/P_m = \mathrm{LG}(m, 2m)$, $L_\omega = \mathcal{O}(1)$, and $W\omega = \{\pm 1\}^m \cong W/W_P$. For a sign vector μ let w_μ be the minimal coset representative with $w_\mu \omega = \mu$, $X_\mu = \overline{Bw_\mu P/P}$, and $\kappa(\mu) = \{m - i + 1 : \mu_i = -1\}$. By Lemma 3.1, $\mu \mapsto \kappa(\mu)$ is a bijection onto strict partitions with parts at most m and $\dim X_\mu = |\kappa(\mu)|$; we write X_κ, w_κ. For Spin_{2m+1} ($\alpha_m = \varepsilon_m$, $\omega_m = \frac{1}{2}\sum \varepsilon_i$) one gets $\mathrm{OG}(m, 2m+1)$ with

the spinor bundle and X_κ^{OG} in the same way. H_κ is a polynomial of degree $|\kappa|$ [11], and $H_\varnothing \equiv 1$. The Demazure operator is $D_i f = (f - e^{-\alpha_i} s_i f)/(1 - e^{-\alpha_i})$, and $D_w = D_{i_1} \cdots D_{i_k}$ for a reduced word. We use two classical inputs and nothing else on the LG side:

(G1) (Demazure character formula [8], see [2, Ch. 3].) For $w \in W^P$ and dominant λ with $W_P \subseteq \mathrm{Stab}(\lambda)$, $H^{i>0}(X_w, L_\lambda) = 0$ and $\mathrm{char}\, H^0(X_w, L_\lambda)^* = D_w(e^\lambda)$.

(LS) (Littelmann [14, §5.2], see [15, p. 305].) $D_w(e^\lambda) = \sum_\pi e^{\pi(1)}$, over LS paths π of shape λ with initial direction $\leq w$.[2]

Lemma 2.2 (Polynomial extension)**.** *For $w \in W$, $P_w(\lambda) = [D_w e^\lambda](0)$, computed on analytic germs $e^\mu(h) = e^{\langle \mu, h\rangle}$ at $h = 0$, is a polynomial in $\lambda \in \mathbb{C}^m$. Given (G1), $H_\kappa(j) = P_{w_\kappa}(j\omega)$ for all $j \in \mathbb{C}$.*

Proof. Let $\mathfrak{m}$ be the maximal ideal of germs. Since $D_i f = (f - s_i f)/(1 - e^{-\alpha_i}) + s_i f$ and $(f - s_i f)/\alpha_i \in \mathfrak{m}^{k-1}$ for $f \in \mathfrak{m}^k$, we get $D_w(\mathfrak{m}^{\ell(w)+1}) \subseteq \mathfrak{m}$; expand e^λ to order $\ell(w)$. For integral λ the value at 0 is the coefficient sum, which is $H_\kappa(j)$ for $j \geq 0$ by (G1); both sides are polynomials in j. □

Lemma 2.3 (The anchor)**.** *If $\kappa \neq \varnothing$, then $H_\kappa(-1) = 0$, both on* LG *and on* OG*.*

Proof. $w_\kappa \neq e$ is minimal in its coset, so its only right descent is s_m: $w_\kappa = w' s_m$ with lengths adding, and $D_{w_\kappa} = D_{w'} D_m$. Since $\langle -\omega_m, \alpha_m^\vee \rangle = -1$, $D_m e^{-\omega_m} = 0$. Apply Lemma 2.2. □

3 The LS-path proof

This section proves Theorem 1.1 from (G1) and (LS) alone. For $\sigma > \tau$ in $W^\lambda = W/W_\lambda$ and rational $a \in (0,1)$, an *a-chain* is a chain of Bruhat covers $\sigma = \sigma_0 > \cdots > \sigma_r = \tau$, $\sigma_{k-1} = s_{\beta_k} \sigma_k$, with $a\langle \sigma_k(\lambda), \beta_k^\vee \rangle \in \mathbb{Z}$ for all k; the number $|\langle \sigma_k(\lambda), \beta_k^\vee \rangle|$ is the *bond* of the cover. An *LS path* of shape λ is a pair $(\tau_1 > \cdots > \tau_s;\ 0 < a_1 < \cdots < a_{s-1} < 1)$ such that an a_i-chain for (τ_i, τ_{i+1}) exists for every i [14, 15]; its initial direction is τ_1.

Lemma 3.1. *In type C_m with $\lambda = \omega_m$:*

(a) $\dim X_\mu = \#\{\alpha > 0 : \langle \mu, \alpha^\vee \rangle < 0\} = \sum_{i:\mu_i=-1}(m - i + 1) = |\kappa(\mu)|$;

(b) the Bruhat covers $\mu < s_\beta \mu$ in W^λ are exactly the moves for which $D(\kappa(s_\beta \mu))$ is $D(\kappa(\mu))$ plus one box;

(c) the bond of such a cover is 1 *if the added box is diagonal and* 2 *otherwise.*

[2] Read first-hand in [15]; the locator in [14] is quoted from Chirivì, Fang and Littelmann (arXiv:2207.08904).

In particular, the Bruhat order on W^λ is inclusion of shifted diagrams.

Sketch. (a) is the inversion count of w_μ, using that ω is dominant and w_μ is minimal. (b,c) Let $i < k$, $p = m - i + 1 > q = m - k + 1$. Upward, $s_{\varepsilon_i - \varepsilon_k}$ replaces the part q by p; it is a cover iff $p = q + 1$, which adds an off-diagonal box at the end of the row of length q, with bond $|1 - (-1)| = 2$. Upward, $s_{\varepsilon_i + \varepsilon_k}$ adds two parts $p + q \geq 3$, never a cover. Upward, $s_{2\varepsilon_i}$ adds the part p, a cover iff $p = 1$: a new diagonal box, with bond $|\langle \mu, \varepsilon_m \rangle| = 1$. □

Lemma 3.2. *Let $j \geq 1$. A pair $(\tau_1 \supsetneq \cdots \supsetneq \tau_s\,;\, 0 < a_1 < \cdots < a_{s-1} < 1)$ of shifted diagrams and rationals is an LS path of shape $j\omega_m$ iff, for every $i \leq s - 1$, $2ja_i \in \mathbb{Z}$, and if $2ja_i$ is odd, then $D(\tau_i) \setminus D(\tau_{i+1})$ contains no diagonal box.*

Proof. $W_{j\omega} = W_\omega$ and all bonds scale by j, so a cover of bond b qualifies iff $jab \in \mathbb{Z}$. Any a-chain has a cover, so $2ja \in \mathbb{Z}$. If $ja \in \mathbb{Z}$, every saturated chain qualifies. If $2ja$ is odd, exactly the bond-2 (off-diagonal) covers qualify, and a saturated chain from τ_{i+1} to τ_i avoiding diagonal boxes exists iff $D(\tau_i) \setminus D(\tau_{i+1})$ has none, since every saturated chain adds every box of the difference. □

Proposition 3.3 (The LS-parity model)**.** *For $j \geq 0$, the LS paths of shape $j\omega_m$ with initial direction $\leq \kappa$ are in bijection with the order-preserving maps $G\colon P_\kappa \to \{0, \ldots, 2j\}$ that are even on diagonal boxes. Hence, given (G1) and (LS), $H_\kappa(j) = h^0(X_\kappa, \mathcal{O}(j)) = F_\kappa(2j)$ for all $j \geq 0$.*

Proof. For $j = 0$ both sides are 1. Let $j \geq 1$. Send a path π to $g(x) = 2j\, a_{i(x)}$, with $a_s = 1$, where $i(x) = \max\{i : x \in D(\tau_i)\}$ (and $g(x) = 0$ if $x \notin D(\tau_1)$). Since the $D(\tau_i)$ are order ideals, g is order-reversing; it is integer-valued and even on diagonal boxes by Lemma 3.2. Conversely, given such g with distinct values $v_1 < \cdots < v_r$ in $(0, 2j)$, put $a_i = v_i/2j$, $\tau_i = \{g \geq v_i\}$ and $\tau_{r+1} = g^{-1}(2j)$; these are shifted diagrams by §2, and Lemma 3.2 makes this an LS path. The maps are inverse, and $G = 2j - g$. The count is (G1)+(LS), and $h^0 = \chi$ by (G1). □

Proof of Theorem 1.1. Rescaling off-diagonal coordinates by 2 maps Λ onto $\mathbb{Z}^{P_\kappa}$ and $\mathcal{O} = \mathcal{O}(P_\kappa)$ onto a lattice polytope, so $L(j) = \#(j\mathcal{O} \cap \Lambda)$ is a polynomial of degree $|\kappa|$. The map $G \mapsto G/2$ is a bijection from the maps counted by $F_\kappa(2j)$ onto $j\mathcal{O} \cap \Lambda$, since $G(x)/2 \in \mathbb{Z}$ iff $G(x)$ is even. By Proposition 3.3, H_κ and L agree at all $j \geq 0$, so $H_\kappa = L$: this is (a) and (c). Under $y = f/2$, the points of $\operatorname{int}(j\mathcal{O}) \cap \Lambda$ are the strictly order-preserving $f\colon P_\kappa \to [2j - 1]$ with even diagonal values. Ehrhart–Macdonald reciprocity gives $H_\kappa(-j) = (-1)^{|\kappa|} N_\kappa(j)$ for $j \geq 1$, which is (b). □

4 The recursion and the Pieri coefficients

For a finite poset P and a subset $D \subseteq P$, let $V(P, D)$ be the number of strictly order-preserving maps f from P to the chain $\{e < o\}$ with $f(D) \subseteq \{e\}$. For an up-set θ of P_κ write $V(\theta) = V(\theta, \text{diagonal boxes of } \theta)$.

Lemma 4.1. *$V(P,D) = 0$ if P contains a chain $x < y < z$, or if some $d \in D$ is not minimal in P. Otherwise $V(P,D) = 2^i$, where i is the number of elements of $P \setminus D$ that are comparable to no other element. In P_κ, a diagonal box (c,c) of an up-set θ is non-minimal in θ iff $(c-1,c) \in \theta$.*

Proof. A 3-chain needs three values, and $x < d$ would force $f(x) < e$. Otherwise P has height ≤ 2, non-maximal elements are forced to e, non-minimal ones to o, consistently; isolated elements outside D are free. For the last claim, $(a,b) < (c,c)$ implies $(a,b) \leq (c-1,c)$. □

Lemma 4.2 (Peeling the top two values)**.** *For $j \geq 1$ and every κ,*

$$N_\kappa(j+1) = N_\kappa(j) + \sum_{\rho \subsetneq \kappa} V\big(D(\kappa) \setminus D(\rho)\big)\, N_\rho(j).$$

Proof. Let f be counted by $N_\kappa(j+1)$ and $\theta = f^{-1}\{2j, 2j+1\}$. Then θ is an up-set, so $\theta = D(\kappa) \setminus D(\rho)$ for a unique ρ. The restriction $f|_\theta$ is a $V(\theta)$-map with $e = 2j$, $o = 2j+1$ ($2j$ is the only even value available on the diagonal), and $f|_{D(\rho)}$ is an $N_\rho(j)$-map. Conversely, any such pair glues to a strictly order-preserving map, because values on $D(\rho)$ are $\leq 2j-1 < 2j \leq$ values on θ, and nothing in θ lies below $D(\rho)$. The term $\theta = \varnothing$ gives $N_\kappa(j)$. □

Proposition 4.3. *On* LG*, put $s_x(\kappa) = (-1)^{|\kappa|} H_\kappa(-x)$ for $x \in \mathbb{C}$. Given (G1) and (LS), for every $x \in \mathbb{C}$,*

$$s_{x+1}(\kappa) = s_x(\kappa) + \sum_{\rho \subsetneq \kappa} V\big(D(\kappa) \setminus D(\rho)\big)\, s_x(\rho). \tag{4.1}$$

Proof. By Theorem 1.1(b), $s_j(\kappa) = N_\kappa(j)$ for every integer $j \geq 1$ and every κ, including $\kappa = \varnothing$. So Lemma 4.2 is (4.1) at every integer $x \geq 1$. For fixed κ both sides of (4.1) are polynomials in x (a finite sum with constant coefficients), so they are equal. □

Remark 4.4 (The Pieri coefficients). By Chevalley's formula $[\mathcal{O}_{X^{(1)}}] = 1 - [L^{-1}]$ for the Schubert divisor $X^{(1)}$. Write $[\mathcal{O}_{X_\kappa}] \cdot (1 - [L^{-1}]) = \sum_\rho c_\rho [\mathcal{O}_{X_\rho}]$ in $K(\mathrm{LG})$. Cutting X_κ by a section of L and using the cell decomposition gives $c_\rho \neq 0 \Rightarrow \rho \subsetneq \kappa$ and

$$H_\kappa(j) - H_\kappa(j-1) = \sum_{\rho \subsetneq \kappa} c_\rho\, H_\rho(j). \tag{4.2}$$

In Buch–Ravikumar's indexing by codimension X_κ is $X^{\kappa^\vee}$, and the reflection $R(a,b) = (m-1-b, m-1-a)$ carries their skew shapes $\rho^\vee/\kappa^\vee$ to $\theta = D(\kappa) \setminus D(\rho)$, turning their letters $1' < 1$ into $o > e$ and their rule "no primed letter on the diagonal" into "diagonal boxes take e". In this translation the case $p = 1$ of their KLG-tableau formula [3, Cor. 5.6], which follows from their recursion [3, Cor. 5.4, Thm. 5.5] by induction on $|\theta|$, reads

$$c_\rho = (-1)^{|\theta|-1} V(\theta).$$

Substituted into (4.2) this is (4.1) again, now with the odd values of Theorem 1.1(b) as the primed letters. For example, for $\kappa = (3,2)$,

$$[\mathcal{O}_{X_{(3,2)}}] \cdot [\mathcal{O}_{X^{(1)}}] = 2[\mathcal{O}_{X_{(3,1)}}] - [\mathcal{O}_{X_{(2,1)}}] - [\mathcal{O}_{X_{(3)}}] + [\mathcal{O}_{X_{(2)}}],$$

with $\theta = \{(1,2)\}$ (isolated off-diagonal box, $V = 2$), $\{(0,2),(1,2)\}$, $\{(1,1),(1,2)\}$, $\{(0,2),(1,1),(1,2)\}$ ($V = 1$ each), and $c_{(1)} = c_{\varnothing} = 0$ because of the 3-chain $(0,1) < (0,2) < (1,2)$. Conversely, (4.1) and (4.2) alone do not determine the c_ρ: for $\kappa = (3,1)$ there are five $\rho \subsetneq \kappa$, all with $\deg H_\rho \leq 3$, so the H_ρ are linearly dependent. Equivariant K-Chevalley formulas [13] determine the c_ρ in principle; we plan to give a T-equivariant proof of the displayed formula that avoids [3] in the full version.

Remark 4.5 (The orthogonal case). On $\mathrm{OG}(m, 2m+1)$, (4.2) holds with the OG coefficients, and [3, Cor. 4.4, Lemma 4.5] give $c_\rho \neq 0$ iff θ is a nonempty antichain of P_κ (equivalently, a set of maximal boxes), and then $c_\rho = (-1)^{|\theta|-1}$. Peeling the top value off a strictly order-preserving map $P_\kappa \to [n+1]$, whose preimage is such an antichain, gives the same recursion for $\overline{\Omega}(P_\kappa, j-1)$, with anchor $\overline{\Omega}(P,0) = \delta_{P,\varnothing} = s_1$ (Lemma 2.3). This proves the second statement of Theorem 1.3 given [3], and Stanley reciprocity gives the first.

5 The integer window

Proof of Theorem 1.4. On LG, $s_j(\kappa) = N_\kappa(j)$ by Theorem 1.1(b); on OG, $s_j(\kappa) = \overline{\Omega}(P_\kappa, j-1)$ by Theorem 1.3. Apply Lemma 2.1. For $\kappa = \varnothing$, $s_j \equiv 1$ and $W = 0$. □

Remark 5.1 (An inductive proof). On LG the window is forced by the shape of (4.1) alone, with no counting formula for s_j. By strong induction on $|\kappa|$: the coefficients of (4.1) are ≥ 0 and $s_1 = 0$ (Lemma 2.3), so $s_j(\kappa)$ is ≥ 0 and weakly increasing. If $V(\theta) \neq 0$, then θ has no 3-chain (Lemma 4.1), so $r(\rho) \geq r(\kappa) - 2$ and $W(\rho) \geq W(\kappa) - 1$; by induction every term of (4.1) vanishes for $j \leq W - 1$, so $s_W = s_1 = 0$. At $j = W$ one term survives: for $T = \{(a,b) \in D(\kappa) : a + b \geq r - t\}$ with $t = 1$ (r odd) or $t = 2$ (r even), T lies on at most two antidiagonals and its diagonal boxes lie on the inner one, so $V(T) \neq 0$, while the corresponding ρ has $W(\rho) < W$. (With the other parity, $V(T)$ can vanish.) The zero anchor is what makes (4.1) monotone at integers; at half-integers the anchor $2^{-\kappa_1}$ is not zero, and §6 needs a finer argument. For $\kappa = (3,2)$, Remark 4.4 gives $s_3(3,2) = s_2(3,2) + 2s_2(3,1) + s_2(2,1) + s_2(3) + s_2(2) = 0+0+0+0+1$ and $s_4(3,2) = 1 + 2\cdot 2 + 1 + 3 + 4 = 13$.

6 Half-integer values

Throughout, $X = \mathrm{LG}(m, 2m)$ and $y_i = e^{-\varepsilon_i}$. Since (4.1) is an identity of polynomials, it holds at half-integers. This section uses only (4.1) (hence (G1) and (LS)), Demazure computations and combinatorics; it does not use [3] or Chevalley's formula.

Proposition 6.1 (Bridge). *For every integer $a \geq 0$, $2^m H_\kappa(-a - \frac{1}{2}) = \left[D_{w_\kappa}(F_a)\right]_{y=1}$, where $F_a = \prod_{i=1}^m y_i^a(1 + y_i)$.*

Sketch. Let $\Phi = \prod_i (1 + y_i)^{-1}$, $D_i' = D_i$ for $i < m$ and $D_m' = (1 + y_m)^{-1} D_m^B$ with $D_m^B g = (g - y_m s_m g)/(1 - y_m)$. Since $e^{-\alpha_m} = y_m^2$, a direct computation gives $D_i(e^{-\omega/2} g) = e^{-\omega/2} D_i' g$ and $D_i'(\Phi g) = \Phi D_i g$, hence $D_w(e^{-\omega/2} g) = e^{-\omega/2} \Phi\, D_w(\Phi^{-1} g)$. Apply this to $g = e^{-a\omega}$ in Lemma 2.2, with $\Phi^{-1} e^{-a\omega} = F_a$ and $(e^{-\omega/2}\Phi)(0) = 2^{-m}$. □

Since F_a is the character of $\mathbb{C}_{-a\omega} \otimes \bigwedge^\bullet U_0$, U_0 of weights $-\varepsilon_i$, (G1) for χ at all weights turns this into the identity $2^m H_\kappa(-a - \frac{1}{2}) = \chi(X_\kappa, \bigwedge^\bullet U \otimes \mathcal{O}(-a))$ quoted in §1; we do not use it.

Theorem 6.2. *$H_\kappa(-\frac{1}{2}) = 2^{-\kappa_1}$ for every κ (with $\kappa_1 = 0$ for $\kappa = \varnothing$).*

Proof. Let $p = m - \kappa_1 + 1$ and $c(k) = s_{m-k+1} \cdots s_m$. The word $c(\kappa_\ell) \cdots c(\kappa_2) c(\kappa_1)$ has length $|\kappa|$ and maps ω to $w_\kappa \omega$, so it is a reduced word for w_κ; its letters left of $c(\kappa_1)$ are s_j with $j \geq m - \kappa_2 + 1 \geq p + 1$. Applying $D_m, \ldots, D_p$ to $\prod_{i \leq m}(1 + y_i)$: before D_q the function is $\prod_{i \leq q}(1 + y_i)$, and D_q fixes $\prod_{i<q}(1 + y_i)$ and kills y_q (pairing -1). So $D_{c(\kappa_1)} F_0 = \prod_{i<p}(1 + y_i)$, which the remaining letters fix. At $y = 1$ this is $2^{m-\kappa_1}$. □

Let $A_o(\kappa, \nu)$ (resp. $A_e(\kappa, \nu)$) be 1 if $\nu \subseteq \kappa$ and $\theta = D(\kappa) \setminus D(\nu)$ is an antichain without diagonal boxes (resp. an antichain), and 0 otherwise.

Lemma 6.3 (Layers). *(a) $(A_o A_e)(\kappa, \nu) = V(D(\kappa) \setminus D(\nu))$. Hence (4.1) reads $s_{x+1} = A_o A_e\, s_x$ for every $x \in \mathbb{C}$.*

(b) For $a \geq 1$, $[(A_o A_e)^{a-1} A_o](\kappa, \nu) = N_{\kappa/\nu}(a)$.

(c) $A_e\, s_{1/2} =: \varepsilon$ has $\varepsilon(\varnothing) = 1$, $\varepsilon((L)) = (-1)^{L+1} 2^{-L}$, and $\varepsilon(\kappa) = 0$ if $\ell(\kappa) \geq 2$.

Proof. (a,b) Peel values from the top as in Lemma 4.2: the preimage of the current top value is an antichain up-set of what remains, without diagonal boxes if the value is odd. (c) By Theorem 6.2, $s_{1/2}(\nu) = (-1)^{|\nu|} 2^{-\nu_1}$. If $\ell(\kappa) \geq 2$, toggling the last box of the bottom row (maximal, not in row 0) cancels terms in pairs; $\kappa = (L)$ has two. □

Corollary 6.4 (Row formula). *For $a \geq 1$, $s_{a+1/2}(\kappa) = N_\kappa(a) + \sum_{L=1}^{\kappa_1} (-1)^{L+1} 2^{-L} N_{\kappa/(L)}(a)$.*

Indeed $s_{a+1/2} = (A_o A_e)^a s_{1/2} = (A_o A_e)^{a-1} A_o\, \varepsilon$. For example, $s_{5/2}(3) = 0 + \frac{3}{2} - \frac{3}{4} + \frac{1}{8} = \frac{7}{8}$, and $H_{(3)}(-\frac{5}{2}) = -\frac{7}{8}$.

Proof of Theorem 1.5. The value at $-\frac{1}{2}$ is Theorem 6.2. Let $a \geq 1$, $M = 2a-1$, $n = \kappa_1$. Fix a map g on rows ≥ 1 counted by $N_{\kappa'}(a)$. Row 0 is a chain of n boxes, and box $(0,\beta)$ lies below $(1,\beta)$; put $u_\beta = g(1,\beta)$ when that box exists and $u_\beta = M+1$ otherwise. Since $g(1,1)$ is even, $u_\beta \geq \beta+1$. Let $G_L(M;u)$ count strictly increasing $v_L < \cdots < v_{n-1}$ in $[M]$ with $v_\beta < u_\beta$ ($G_n = 1$), let E_0 be the same count for $L = 0$ with v_0 even, and $\Phi(u) = E_0 - \sum_{L=1}^{n}(-\frac{1}{2})^L G_L$. Summing Corollary 6.4 over g gives $s_{a+1/2}(\kappa) = \sum_g \Phi(u(g))$. Put $\Psi_L = \sum_{L' \geq L}(-\frac{1}{2})^{L'-L} G_{L'}$, so $\Phi = E_0 + \frac{1}{2}\Psi_1$. Splitting by whether $v_L = 1$ and deleting that box gives the *shift* $\Psi_L(M;u) = G_L(M-1;u-1) + \frac{1}{2}\Psi_{L+1}(M-1;u-1)$, valid while $M \geq 1$ and $u \geq 2$, which $u_\beta \geq \beta+1$ guarantees along the iteration. If $2a \leq n$, then $M \leq n-1$, every G-term and E_0 vanish for lack of values, and $\Phi \equiv (-1)^{n-1-M}2^{-n} = (-1)^{\kappa_1}2^{-\kappa_1}$. If $2a \geq n$, the iteration reaches $\Psi_n = 1$ through nonnegative terms, so $\Phi \geq 2^{-n}$. Hence $s_{a+1/2}(\kappa) = (-1)^{\kappa_1}2^{-\kappa_1}N_{\kappa'}(a)$, resp. $s_{a+1/2}(\kappa) \geq 2^{-\kappa_1}N_{\kappa'}(a)$, and it vanishes iff $N_{\kappa'}(a) = 0$, i.e. by Lemma 2.1 iff $\kappa' \neq \varnothing$ and $a \leq \lceil r(\kappa')/2 \rceil$. Signs follow from $H_\kappa = (-1)^{|\kappa|}s$, and $\Phi \in 2^{-n}\mathbb{Z}$. □

Proof of Corollary 1.6. Let $M = 2a+1 \geq 2m+1$, so $2a \geq \kappa_1$. As $\kappa'_1 \leq m-1$, $r(\kappa') \leq 2m-3$ and $\lceil r(\kappa')/2 \rceil < a$. By Theorem 1.5, $2^{\kappa_1}s_{a+1/2}(\kappa)$ is a positive integer, and so is $2^m s_{a+1/2}(\kappa)$. □

Remark 6.5 (Vanishing by hyperplane sections). The vanishing half of Theorem 1.5(i) has a second proof that uses only the *support* of (4.1). A Demazure computation like that for Theorem 6.2 gives $H_\kappa(-\frac{3}{2}) = 2^{-\kappa_1}H_{\kappa'}(-1)$ for $\kappa_1 \geq 2$, which is 0 when $\ell(\kappa) \geq 2$. If $V(\theta) \neq 0$, then $r(\rho') \geq r(\kappa') - 2$, so by induction every term of $s_{x+1} - s_x$ vanishes at $x = \frac{3}{2}, \frac{5}{2}, \ldots$ inside the window, and the zero at $-\frac{3}{2}$ propagates. The terms at the edge of the window do not share a sign (for $\kappa = (3,2)$ they are $-\frac{1}{8}$ and $+\frac{1}{4}$), so nonvanishing needs the fibre argument above.

Remark 6.6 (The half-step operator). The unitriangular square root T of $A_o A_e$ satisfies $s_{x+1/2} = T\, s_x$, and $T(\kappa,\nu)$ is a product over the components D of θ of weights 1 (single off-diagonal box), $\pm(-\frac{1}{2})^{|D|}$ (ribbon with one diagonal box) or 0 (full version).

7 Example and open questions

Table 1 was computed from Demazure characters, independently of the theorems. For $\kappa = (3)$: $s_3 = 3$ counts the triples $2 \leq x < y < z \leq 5$ with x even, and at $-\frac{3}{2}$ we have $2a = 2 \leq 3$, so $H = 2^{-3}N_\varnothing = \frac{1}{8}$. For $\kappa = (3,2,1)$, $X_\kappa = \mathrm{LG}(3,6)$: $r = 5$ gives the integer zeros $-1,-2,-3$, and $\kappa' = (2,1)$ with $r' = 3$ gives the half-integer zeros $-\frac{3}{2}, -\frac{5}{2}$; then $H(-\frac{7}{2}) = \frac{1}{8}$ attains the bound of Theorem 1.5(iii). Computer checks: Theorems 1.1(b) and 1.3 against Demazure characters for all κ with $m \leq 5$, $j \leq 4$; Theorem 1.5 for $m \leq 4$, $a \leq 5$; Theorem 1.4 independently by Clio Vega for $m \leq 6$ on both LG and OG.

κ	-1	-2	-3	-4	-5	$-1/2$	$-3/2$	$-5/2$	$-7/2$	$-9/2$
(1)	0	-1	-2	-3	-4	$1/2$	$-1/2$	$-3/2$	$-5/2$	$-7/2$
(2)	0	1	4	9	16	$1/4$	$1/4$	$9/4$	$25/4$	$49/4$
(3)	0	0	-3	-13	-34	$1/8$	$1/8$	$-7/8$	$-55/8$	$-175/8$
$(2,1)$	0	0	-1	-5	-14	$1/4$	0	$-1/4$	$-5/2$	$-35/4$
$(3,1)$	0	0	2	15	56	$1/8$	0	$3/8$	$25/4$	$245/8$
$(3,2)$	0	0	-1	-13	-70	$1/8$	0	$-1/8$	$-17/4$	$-259/8$
$(3,2,1)$	0	0	0	1	14	$1/8$	0	0	$1/8$	$35/8$

Table 1: Values of H_κ at negative integers and half-integers on $\mathrm{LG}(3,6)$.

Question 7.1 (Other cominuscule spaces)**.** The cominuscule G/P that are not minuscule are $\mathrm{LG}(m,2m)$ and the odd quadrics $Q^{2m-1} = B_m/P_1$. For the odd quadric the Bruhat poset is a chain, and exactly one cover has bond 2; this is the lecture-hall setting of [5]. Is the Hilbert function of every Schubert variety in a cominuscule G/P the Ehrhart polynomial of its order polytope with respect to the lattice determined by the bonds, as in Theorem 1.1(c)? Is there a uniform reason, independent of the case-by-case bond computation of Lemma 3.1?

Question 7.2 (Higher Pieri coefficients)**.** Do the coefficients $c^{\nu}_{\lambda,p}$ of [3] admit an interpretation by strict maps into a $2p$-letter alphabet $\{e_1 < o_1 < \cdots < e_p < o_p\}$ with even letters on the diagonal, extending Remark 4.4?

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AI declaration

The mathematics in this extended abstract was produced by an AI agent. "Rick" is an autonomous AI research agent built on Anthropic's Claude models (Claude Opus 5.5 in the final sessions; earlier sessions ran on earlier Claude versions **[Robin:** confirm the list]). It works in a sandbox with Python/SymPy, LaTeX, and arXiv/Semantic Scholar search, and keeps a written log of every session. Rick chose the question, ran the computer experiments, conjectured the statements, found and wrote the proofs, did the literature searches, and drafted this text.

- **Proved here from (G1) and (LS) only** (no use of [3] or Chevalley's formula): Lemma 3.1, Lemma 3.2, Proposition 3.3, Theorem 1.1, Lemmas 4.1 and 4.2, the recursion (4.1) (Proposition 4.3), Theorem 1.4 on LG and the inductive argument of Remark 5.1, Proposition 6.1, Theorem 6.2, Lemma 6.3, Corollary 6.4, Theorem 1.5 and Corollary 1.6, and the arguments of Remarks 6.5 and 6.6. Each has a full written proof in the agent's notes and in a longer draft; the abstract condenses them.

- **Cited from Buch–Ravikumar [3]:** the values $c_\rho = (-1)^{|\theta|-1}V(\theta)$ in Remark 4.4 (the case $p = 1$ of [3, Cor. 5.6], which the agent derived from [3, Cor. 5.4, Thm. 5.5] and checked line by line against their statements, and which was also checked by machine against their eq. (11) for all shapes at $m \leq 7$), and the OG coefficients in Remark 4.5 [3, Cor. 4.4, Lemma 4.5]. The translation between indexings was proved by the agent (a T-fixed point and dimension count). Nothing in the main line depends on these remarks.

- **Cited classical results:** Theorem 1.3 (minuscule standard monomial theory and Stanley reciprocity [12, 18, 9, 10, 19, 21]; these references were not read first-hand for this abstract, and nothing on the LG side depends on them), the Demazure character formula (G1), Littelmann's path model (LS), Ehrhart–Macdonald reciprocity, and polynomiality of Euler characteristics [11]. The first-hand reading of [14] is limited as stated in the footnote to (LS).

- **Review by a second AI agent.** Clio Vega, a separate AI research agent (also Claude-based, run independently), line-read and endorsed the inductive proof of Theorem 1.4 on LG and OG and the support-only vanishing argument of Remark 6.5, and independently checked Theorem 1.4 for $m \leq 6$ and the $p = 1$ Pieri values on 1243 shapes. Theorem 1.1, the recursion (4.1), Theorem 1.5, Corollary 1.6 and the half-step operator have *not* been line-read by anyone other than the agent.

- **Computer verification** is listed in §7. Sub-agents (further instances of the same model) ran some computations; their outputs were checked by the agent.

- **Human role.** **[Robin:** describe your role (supervision, submission, any human verification). As of this draft no proof has been checked by a human.**]**